Type MS · Using a binomial expansion to estimate a value

Answer 1 of 12 · Question 9 part b · 3 marks

2020 JAN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

Number		
9 (a)	$1+\frac{1}{2}(-4x)+\frac{\frac{1}{2}\left(-\frac{1}{2}\right)(-4x)^2}{2!}+\frac{\frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)(-4x)^3}{3!}$	M1
	$1-2x-2x^2-4x^3$	A1 A1 (3)
(b)	$x=0.06$ $1-0.12-0.0072-0.000864$ 0.8719	B1 M1 A1 (3)
(c)	$\sqrt{\frac{76}{100}}=\frac{1}{5}\sqrt{19}$ $\sqrt{19}=0.8719\times 5$ 4.360	M1 A1 (2)
		[8]

Part	Mark	Additional Guidance
(a)	M1	For an attempt at a Binomial expansion. A attempt is defined as the following • The expansion must start with 1 • The powers of x must be correct • $-4x$ must be used at least once • The denominators (2! And 3!) must be seen. Accept 2 and 6 $(1-4x)^{\frac{1}{2}}=1+\left(\frac{1}{2}\right)(-4x)+\frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)}{2!}(-4x)^2+\frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{3!}(-4x)^3$
	A1	For at least one term in x correct and fully simplified. $1-2x-2x^2-4x^3$
	A1	For the expansion fully correct and simplified
(b)	B1	For finding the value of $x=0.06$
	M1	For substituting **their** value of x into the expansion **provided** $\lvert x\rvert \leqslant 0.25$ Use of **their expansion** or the correct expansion must be seen explicitly here
	A1	0.8719
(c)	M1	For using their value from (b) in $\sqrt{0.76}=\frac{\sqrt{19}}{5}\Rightarrow\sqrt{19}=5\sqrt{0.76}=5\times\text{'0.8719'}$
	A1	For 4.360 rounded correctly
Penalise rounding once only in this question. Answers must round to the given answers.		

Type MS · Using a binomial expansion to estimate a value

Answer 1 of 12 · Question 9 part b · 3 marks

2020 JAN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Using a binomial expansion to estimate a value

Answer 2 of 12 · Question 9 part d · 4 marks

2022 JAN · Paper 01 · extract 1 of 3 · find part d in this mark scheme extract

9 (a)	$(1-2x)^{-\frac{1}{2}} = 1+\left(-\frac{1}{2}\times -2x\right)+\frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)(-2x)^2}{2!}+\frac{\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{5}{2}\right)(-2x)^3}{3!}+\ldots$ $(1-2x)^{-\frac{1}{2}} = 1+x+\frac{3}{2}x^2+\frac{5}{2}x^3+\ldots$.	M1 A1A1 [3]
(b)	$\frac{1}{\sqrt{0.96}} = \frac{1}{\sqrt{\frac{96}{100}}} = \frac{10}{4\sqrt{6}}\times\frac{\sqrt{6}}{\sqrt{6}} = \ldots$ $\frac{10\sqrt{6}}{4\times 6} = \frac{5\sqrt{6}}{12}$ *	M1 A1 cso [2]
ALT – confirming given result		
	$\frac{1}{\sqrt{0.96}} = \frac{5\sqrt{6}}{12} \Rightarrow 12 = \sqrt{0.96}\times 5\sqrt{6}$ $12^2 = 0.96\left(5\sqrt{6}\right)^2 = 0.96\times 5^2\times 6$ *	[M1 A1cso]
(c)	$\frac{1}{\left(5\sqrt{6}-12\right)}\times\frac{\left(5\sqrt{6}+12\right)}{\left(5\sqrt{6}+12\right)}$ $= \frac{5\sqrt{6}+12}{150-12^2} = \frac{5\sqrt{6}+12}{6} = \frac{5\sqrt{6}}{6}+2$	M1 A1 [2]
(d)	$1-2x = 0.96 \Rightarrow 2x = 0.04 \Rightarrow x = 0.02$ $\frac{9}{5\sqrt{6}-12} = 9\left(2\times\left[\frac{5\sqrt{6}}{12}\right]+2\right) =: 9\times\left[2\left(1+0.02+\frac{3}{2}\times 0.02^2+\frac{5}{2}\times 0.02^3\right)+2\right] = \ldots$ 36.37116	B1 M1:M1 A1 [4]
		Total 11 marks

Type MS · Using a binomial expansion to estimate a value

Answer 2 of 12 · Question 9 part d · 4 marks

2022 JAN · Paper 01 · extract 2 of 3 · find part d in this mark scheme extract

		$(1-2x)^{-\frac{1}{2}} = 1 + x + \frac{3}{2}x^2 + \frac{5}{2}x^3 + \ldots$
	A1	Fully correct simplified expansion as shown above.
(b)	**M1**	For changing 0.96 to $\frac{96}{100}$ or equivalent fraction and attempting to multiply numerator and denominator by either $\sqrt{6}$ $\left(\text{or } \sqrt{96}\right)$ $\frac{1}{\sqrt{0.96}} = \frac{1}{\sqrt{\frac{96}{100}}} = \frac{10}{4\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}} = \ldots$
	A1 cso	For the correct answer as shown with no errors $\frac{10\sqrt{6}}{4\times 6} = \frac{5\sqrt{6}}{12}*$
ALT – confirming given result		
	M1	For rearranging and squaring OR for squaring on both sides
	A1 cso	For showing that the two sides of the result are equal $12^2 = 0.96\left(5\sqrt{6}\right)^2 = 0.96 \times 5^2 \times 6 = 144$
(c)	**M1**	For multiplying numerator and denominator by $5\sqrt{6}+12$ $\frac{1}{\left(5\sqrt{6}-12\right)} \times \frac{\left(5\sqrt{6}+12\right)}{\left(5\sqrt{6}+12\right)}$ [Can be implied by $\frac{\left(5\sqrt{6}+12\right)}{\left(150-12^2\right)}$ seen]
	A1	For a **correct** expansion of brackets throughout. $\frac{1}{(5\sqrt{6}-12)} \times \frac{(5\sqrt{6}+12)}{(5\sqrt{6}+12)}\left[= \frac{5\sqrt{6}+12}{150-12^2} =\right] \frac{5\sqrt{6}+12}{6} = \frac{5\sqrt{6}}{6} + 2$
(d)	**B1**	For finding the required value of x $1-2x = 0.96 \Rightarrow 2x = 0.04 \Rightarrow x = 0.02$
	M1	For substituting their value of x **provided** it is $-\frac{1}{2} < x < \frac{1}{2}$ into the expansion as follows: $\frac{9}{5\sqrt{6}-12} = 9\left(2 \times \left[\frac{5\sqrt{6}}{12}\right] + 2\right) =$ **Note:** Must show substitution if x is incorrect.
	M1	For substituting their expansion for $\frac{5\sqrt{6}}{12}$ $9 \times \left[2\left(1 + 0.02 + \frac{3}{2} \times 0.02^2 + \frac{5}{2} \times 0.02^3\right) + 2\right] = \cdots$
	A1	For the value of 36.37116 [The calculator value is 36.37117]

Type MS · Using a binomial expansion to estimate a value

Answer 2 of 12 · Question 9 part d · 4 marks

2022 JAN · Paper 01 · extract 3 of 3 · find part d in this mark scheme extract

Question	Scheme	Marks

Type MS · Using a binomial expansion to estimate a value

Answer 3 of 12 · Question 6 part c · 2 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

6 a	$p=2 \quad q=\frac{3}{8}$	B1 B1 (2)
b	$(2)\left[1+\frac{1}{3}\left("\frac{3}{8}"x\right)+\frac{\frac{1}{3}\left(-\frac{2}{3}\right)}{2!}\left("\frac{3}{8}"x\right)^2\right]=2+\frac{1}{4}x-\frac{1}{32}x^2$	M1 A1ft A1 (3)
c	$\left(\sqrt[3]{9}=\sqrt[3]{8+1}\Rightarrow x=\frac{1}{3}\right)$	
	So $"2"+"\frac{1}{4}"\times"\frac{1}{3}"-"\frac{1}{32}"\times\left("\frac{1}{3}"\right)^2\left(=\frac{576+24-1}{288}\right)=\frac{599}{288}$ *	M1 A1 cso (2)
		Total 7 marks

Part	Mark	Additional Guidance
(a)	B1	For $p=2$ **or** $q=\frac{3}{8}$ condone $8^{\frac{1}{3}}$
	B1	For $p=2$ **and** $q=\frac{3}{8}$ condone $8^{\frac{1}{3}}$
(b)	M1	For an attempt to expand $(1+qx)^{\frac{1}{3}}$ with their value of q up to the term in x^2 It is not necessary to see p at this stage. The definition of an attempt is as follows: • The first term must be 1 • The next term must be correct for their value of q • The powers of qx must be correct eg $(qx)^2$ • The denominators must be correct Simplification not required. Do not allow missing brackets unless recovered later – this is a general point of marking. Ignore any terms with powers higher than 2.
	A1ft	For both algebraic terms fully correct and unsimplified in the expansion of $(1+qx)^{\frac{1}{3}}$ for their value of q. It is not necessary to see p at this point. Ignore any terms with powers higher than 2.
	A1	For all 3 terms correct, all simplified. Ignore any terms with powers higher than 2
If there are any other methods used – send this to review please.		
(c)	M1	For correct substitution of $x=\frac{1}{3}$ into their expansion, which must have at least 2 terms, including a term in x^2
	A1cso*	Obtains the given result

Type MS · Using a binomial expansion to estimate a value

Answer 3 of 12 · Question 6 part c · 2 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks
	-6 -5 -4 –3 –2 –1 0	B2

Type MS · Using a binomial expansion to estimate a value

Answer 4 of 12 · Question 6 part c · 3 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

6 (a)	$\sqrt{4-x}=2\left(1-\frac{x}{4}\right)^{\frac{1}{2}}$ or $A=2, B=4$	B1B1 [2]
(b)	$2\left\{1+\left(\frac{1}{2}\right)\left(-\frac{x}{4}\right)+\frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{x}{4}\right)^2}{2!}+\frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{x}{4}\right)^3}{3!}+...\right\}$ $=2-\frac{x}{4}-\frac{x^2}{64}-\frac{x^3}{512}+...$	M1A1ft A1 [3]
(c)	$\frac{\sqrt{305}}{10}=\sqrt{4-x}\Rightarrow x=0.95$ $2-\frac{0.95}{4}-\frac{0.95^2}{64}-\frac{0.95^3}{512}=1.7467238.....\approx 1.74672$	B1 M1A1 [3]
		Total 8 marks

Part	Mark	Additional Guidance
(a)	B1B1	B1B0 for one correct value, B1B1 For the correct values of both A and B A and B must be integers from correct working **Do not accept** $A=4^{\frac{1}{2}}$ or $\sqrt{4}$
(b)	M1	Attempt to expand their $'2'\left(1-\frac{x}{'4'}\right)^{\frac{1}{2}}$ up to x^3 term (with or without '2') Must have: • First term is 1 for their $\left(1-\frac{x}{'4'}\right)^{\frac{1}{2}}$ expansion • Correct powers of $\pm\frac{x}{'4'}$ for each term No simplification needed. Ignore terms beyond x^3
	A1ft	Two correct algebraic terms in their expansion, with or without'2' Not necessarily in lowest terms. Ignore terms beyond x^3.
	A1	All terms correct and in lowest terms. Ignore terms beyond x^3.
(c)	B1	Identifies $x=0.95$ oe. e.g. $\frac{19}{20}$
	M1	Substitutes their x which must come from some working (provided $\lvert x\rvert<4$) into each of the algebraic terms of their expansion.
	A1	1.74672 (must be 5dp) **Note:** Calculator value for $\frac{\sqrt{305}}{10}$ is 1.74642492

Type MS · Using a binomial expansion to estimate a value

Answer 4 of 12 · Question 6 part c · 3 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Using a binomial expansion to estimate a value

Answer 5 of 12 · Question 6 part c · 3 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

6 (a)	$\sqrt{9-x}=3\left(1-\frac{x}{9}\right)^{\frac{1}{2}} \Rightarrow p=3, \quad q=-\frac{1}{9}$	B1B1 [2]
(b)	$3\left(1-\frac{x}{9}\right)^{\frac{1}{2}}=$ $3\left\{1+\left(\frac{1}{2}\right)\left(-\frac{x}{9}\right)+\frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{x}{9}\right)^2}{2!}+\frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\left(-\frac{x}{9}\right)^3}{3!}+\ldots\right\}$ $=3\left(1-\frac{x}{18}-\frac{x^2}{648}-\frac{x^3}{11664}\right)=3-\frac{x}{6}-\frac{x^2}{216}-\frac{x^3}{3888}$	M1 A1A1 [3]
(c)	$\frac{31}{4}=9-x \Rightarrow x=\frac{5}{4}$ $\sqrt{\frac{31}{4}} \approx 3-\frac{1.25}{6}-\frac{1.25^2}{216}-\frac{1.25^3}{3888}=2.783930523\ldots \approx 2.78393$	B1 M1A1 [3]
		Total 8 marks

(a)	B1	$p=3$ or $\sqrt{9}$ or $9^{\frac{1}{2}}$	May be shown in their $3\left(1-\frac{x}{9}\right)^{\frac{1}{2}}$ (allow isw)
	B1	$q=-\frac{1}{9}$	
(b)	M1	Attempt to use the binomial expansion for their $(1+qx)^{\frac{1}{2}}$. Must have first term 1, three more terms with ascending powers of x, 2 or 2! and 6 or 3! Seen, and their $\left(-\frac{x}{9}\right)$ used at least once. No simplification needed. Ignore terms beyond x^3	
	A1	Two algebraic terms correct in the expansion of $3\left(1-\frac{x}{9}\right)^{\frac{1}{2}}$. Must be single fractions, not necessarily in lowest terms. Ignore terms beyond x^3	
	A1	All four terms correct and in lowest terms. Ignore terms beyond x^3	
(c)	B1	Identify $x=\frac{5}{4}$ as the value needed to make $\sqrt{9-x}=\sqrt{\frac{31}{4}}$	
	M1	Substitute their x into each of the algebraic terms of their expansion of $\sqrt{9-x}$	
	A1	2.78393 **given to 5DP**.	

Type MS · Using a binomial expansion to estimate a value

Answer 5 of 12 · Question 6 part c · 3 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Using a binomial expansion to estimate a value

Answer 6 of 12 · Question 1 part b · 3 marks

2023 JAN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

1(a)	$\left(1+\frac{x}{4}\right)^8 = 1+2x+\frac{7}{4}x^2+\frac{7}{8}x^3$ oe	B1M1A1 [3]
(b)	$1+\frac{x}{4}=1.035 \Rightarrow x=0.14$ oe $(1.035)^8 = 1+2(0.14)+1.75(0.14)^2+0.875(0.14)^3 = 1.3167$ [NB: the calculator value is 1.316809037]	B1 M1A1 [3]
		Total 6 marks

Question	Notes	Marks
1	$\left(1+\frac{x}{4}\right)^8$	
(a)	For an attempt at the binomial expansion of $\left(1+\frac{x}{4}\right)^8$ $\left(1+\frac{x}{4}\right)^8 = 1+2x+...$	B1
	For either correct third or fourth term. Need not be simplified. $...+C_2^8\times\left(\frac{x}{4}\right)^2$ or $C_3^8\times\left(\frac{x}{4}\right)^3$	M1
	For the correct expansion fully simplified. $\left(1+\frac{x}{4}\right)^8 = 1+2x+\frac{7}{4}x^2+\frac{7}{8}x^3$ OR $\left(1+\frac{x}{4}\right)^8 = 1+2x+1.75x^2+0.875x^3$ Do not ignore subsequent incorrect simplification.	A1 [3]
(b)	Finds the value of x to be substituted. $1+\frac{x}{4}=1.035 \Rightarrow x=0.14$ oe	B1
	Substitutes their value of x into their expansion provided it has at least 2 terms in x. *Their* $x \neq 1.035$ $(1.035)^8 = 1+2(0.14)+1.75(0.14)^2+0.875(0.14)^3$	M1
	For the correct value of $(1.035)^8 = 1.3167$ rounded correctly. [NB: the calculator value is 1.316809037]	A1 [3]
		Total 6 marks

Type MS · Using a binomial expansion to estimate a value

Answer 6 of 12 · Question 1 part b · 3 marks

2023 JAN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Using a binomial expansion to estimate a value

Answer 7 of 12 · Question 9 part c · 2 marks

2020 JAN · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

Number		
9 (a)	$1+\frac{1}{2}(-4x)+\frac{\frac{1}{2}\left(-\frac{1}{2}\right)(-4x)^2}{2!}+\frac{\frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)(-4x)^3}{3!}$	M1
	$1-2x-2x^2-4x^3$	A1 A1 (3)
(b)	$x=0.06$ $1-0.12-0.0072-0.000864$ 0.8719	B1 M1 A1 (3)
(c)	$\sqrt{\frac{76}{100}}=\frac{1}{5}\sqrt{19}$ $\sqrt{19}=0.8719\times 5$ 4.360	M1 A1 (2)
		[8]

Part	Mark	Additional Guidance
(a)	M1	For an attempt at a Binomial expansion. A attempt is defined as the following • The expansion must start with 1 • The powers of x must be correct • $-4x$ must be used at least once • The denominators (2! And 3!) must be seen. Accept 2 and 6 $(1-4x)^{\frac{1}{2}}=1+\left(\frac{1}{2}\right)(-4x)+\frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)}{2!}(-4x)^2+\frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{3!}(-4x)^3$
	A1	For at least one term in x correct and fully simplified. $1-2x-2x^2-4x^3$
	A1	For the expansion fully correct and simplified
(b)	B1	For finding the value of $x=0.06$
	M1	For substituting **their** value of x into the expansion **provided** $\lvert x\rvert \leqslant 0.25$ Use of **their expansion** or the correct expansion must be seen explicitly here
	A1	0.8719
(c)	M1	For using their value from (b) in $\sqrt{0.76}=\frac{\sqrt{19}}{5}\Rightarrow\sqrt{19}=5\sqrt{0.76}=5\times\text{'0.8719'}$
	A1	For 4.360 rounded correctly
Penalise rounding once only in this question. Answers must round to the given answers.		

Type MS · Using a binomial expansion to estimate a value

Answer 7 of 12 · Question 9 part c · 2 marks

2020 JAN · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Using a binomial expansion to estimate a value

Answer 8 of 12 · Question 9 part c.i · 0 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 3 · find part c.i in this mark scheme extract

		For attempt to integrate one of: $9-9e^{-3x}$ or $e^{3x}-1$ or $\pm[(9-9e^{-3x})-(e^{3x}-1)]$ Limits may not be present. At least one term correct. Ignore +c if included.
	A1$_2$	For correct integration of one of the exponential terms $\pm 9e^{-3x} \rightarrow \mp\frac{9}{3}e^{-3x}$ **or** $\pm e^{3x} \rightarrow \pm\frac{1}{3}e^{3x}$ Limits need not be present. Ignore +c if included.
	A1$_3$	For correct integration of both curves $9x+3e^{-3x}$ **and** $\frac{e^{3x}}{3}-x$ or for a **fully correct** integration where the difference between two expressions is found $\pm\left(10x-\frac{9}{-3}e^{-3x}-\frac{e^{3x}}{3}\right)$ or $\pm\left(9x-\frac{9}{-3}e^{-3x}-\frac{e^{3x}}{3}+x\right)$ **Note: this is an M mark in epen**
	M1$_4$	For the difference between the two expressions either before or after integration. Allow subtraction either way around. **Note: this is an A mark in epen**
	dM1$_5$	Substitution of correct limits into their integrated expressions (limits subtracted the correct way around). Dependent on first M scored. If substituting before difference found then must substitute into both integrated expressions. May be implied by awrt 1.99 If integration is not correct then substitution must be shown.
	A1$_6$	For the correct answer oe. Must be exact value.

Type MS · Using a binomial expansion to estimate a value

Answer 8 of 12 · Question 9 part c.i · 0 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 3 · find part c.i in this mark scheme extract

Question number	Scheme	Marks
9(a)	$(3(3-x)^{-3} = \quad \frac{1}{9}\left(1-\frac{x}{3}\right)^{-3}\ )$ $\quad a=\frac{1}{9} \quad b=\frac{1}{3}$	B1 B1 [2]
(b)	$\left(1-\frac{x}{3}\right)^{-3} =$ $\left[1+(-3)\left(-\frac{x}{3}\right)+\frac{(-3)(-4)\left(-\frac{x}{3}\right)^2}{2!}+\frac{(-3)(-4)(-5)\left(-\frac{x}{3}\right)^3}{3!}\right]$	M1 A1ft
	$\frac{1}{9}+\frac{1}{9}x+\frac{2}{27}x^2+\frac{10}{243}x^3$	A1 [3]
(c) (i)	$\frac{24}{125}=\frac{3}{(3-x)^3}$ or $\frac{125}{8}=(3-x)^3 \Rightarrow \frac{5}{2}=3-x$	
	$x=0.5$ oe	B1
	$\frac{1}{9}+\frac{1}{9}("0.5")+\frac{2}{27}("0.5")^2+\frac{10}{243}("0.5")^3 \quad (=0.19033)$	B1ft
(ii)	$\pm\left(\frac{\frac{24}{125}-"0.19033"}{\frac{24}{125}}[\times 100]\right)$ oe	M1
	0.87% or −0.87%	A1 [4]
		Total 9 marks

Type MS · Using a binomial expansion to estimate a value

Answer 8 of 12 · Question 9 part c.i · 0 marks

2022 JUN · Paper 02 (regional R) · extract 3 of 3 · find part c.i in this mark scheme extract

Question number	Scheme	Marks

		For attempt to integrate one of: $9-9e^{-3x}$ or $e^{3x}-1$ or $\pm[(9-9e^{-3x})-(e^{3x}-1)]$ Limits may not be present. At least one term correct. Ignore +c if included.
	$A1_2$	For correct integration of one of the exponential terms $\pm 9e^{-3x} \rightarrow \mp\frac{9}{3}e^{-3x}$ **or** $\pm e^{3x} \rightarrow \pm\frac{1}{3}e^{3x}$ Limits need not be present. Ignore +c if included.
	$A1_3$	For correct integration of both curves $9x+3e^{-3x}$ **and** $\frac{e^{3x}}{3}-x$ or for a **fully correct** integration where the difference between two expressions is found $\pm\left(10x-\frac{9}{-3}e^{-3x}-\frac{e^{3x}}{3}\right)$ or $\pm\left(9x-\frac{9}{-3}e^{-3x}-\frac{e^{3x}}{3}+x\right)$ **Note: this is an M mark in epen**
	$M1_4$	For the difference between the two expressions either before or after integration. Allow subtraction either way around. **Note: this is an A mark in epen**
	$dM1_5$	Substitution of correct limits into their integrated expressions (limits subtracted the correct way around). Dependent on first M scored. If substituting before difference found then must substitute into both integrated expressions. May be implied by awrt 1.99 If integration is not correct then substitution must be shown.
	$A1_6$	For the correct answer oe. Must be exact value.

Type MS · Using a binomial expansion to estimate a value

Answer 9 of 12 · Question 9 part c.ii · 4 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 3 · find part c.ii in this mark scheme extract

Question number	Scheme	Marks
9(a)	$(3(3-x)^{-3} = \quad \frac{1}{9}\left(1-\frac{x}{3}\right)^{-3})$ $\quad a=\frac{1}{9} \quad b=\frac{1}{3}$	B1 B1 [2]
(b)	$\left(1-\frac{x}{3}\right)^{-3} =$ $\left[1+(-3)\left(-\frac{x}{3}\right)+\frac{(-3)(-4)\left(-\frac{x}{3}\right)^2}{2!}+\frac{(-3)(-4)(-5)\left(-\frac{x}{3}\right)^3}{3!}\right]$	M1 A1ft
	$\frac{1}{9}+\frac{1}{9}x+\frac{2}{27}x^2+\frac{10}{243}x^3$	A1 [3]
(c) (i)	$\frac{24}{125}=\frac{3}{(3-x)^3}$ or $\frac{125}{8}=(3-x)^3 \Rightarrow \frac{5}{2}=3-x$	
	$x=0.5$ oe	B1
	$\frac{1}{9}+\frac{1}{9}("0.5")+\frac{2}{27}("0.5")^2+\frac{10}{243}("0.5")^3 \quad (=0.19033)$	B1ft
(ii)	$\pm\left(\frac{\frac{24}{125}-"0.19033"}{\frac{24}{125}}[\times 100]\right)$ oe	M1
	0.87% or −0.87%	A1 [4]
		Total 9 marks

Type MS · Using a binomial expansion to estimate a value

Answer 9 of 12 · Question 9 part c.ii · 4 marks

2022 JUN · Paper 02 (regional R) · extract 3 of 3 · find part c.ii in this mark scheme extract

Question number	Scheme	Marks

Type MS · Using a binomial expansion to estimate a value

Answer 10 of 12 · Question 6 part d · 3 marks

2024 NOV · Paper 01 · find part d in this mark scheme extract

Number		
6(a)	eg $\frac{a}{\sqrt{4\left(1+\frac{bx}{4}\right)}}$ or $\frac{a}{\sqrt{4}\sqrt{\left(1+\frac{bx}{4}\right)}}$ or $\frac{a}{4^{\frac{1}{2}}\left(1+\frac{bx}{4}\right)^{\frac{1}{2}}}$ or $a\left(4^{-\frac{1}{2}}\right)\left(1+\frac{bx}{4}\right)^{-\frac{1}{2}}$ or $a\left(4\left(1+\frac{bx}{4}\right)\right)^{-\frac{1}{2}}$	M1
	$=\frac{a}{2}\left(1+\frac{bx}{4}\right)^{-\frac{1}{2}}$	A1*cso [2]
(b)	$\left(\frac{a}{2}\right)\left[1+\left(-\frac{1}{2}\right)\left(\frac{bx}{4}\right)+\frac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)}{2!}\left(\frac{bx}{4}\right)^2+\frac{\left(-\frac{1}{2}\right)\left(-\frac{1}{2}-1\right)\left(-\frac{1}{2}-2\right)}{3!}\left(\frac{bx}{4}\right)^3\right]$ oe	M1 M1 A1
	$\left(\left(\frac{a}{2}\right)\left[1+\left(-\frac{bx}{8}\right)+\frac{3}{128}b^2x^2+\left(-\frac{5}{1024}b^3x^3\right)\right]\right)$ Not necessary - but must be checked if seen All intermediate steps must be checked for errors	
	$\frac{a}{2}-\frac{ab}{16}x+\frac{3ab^2}{256}x^2-\frac{5ab^3}{2048}x^3$ $P=\frac{a}{2},Q=-\frac{ab}{16}*,R=\frac{3ab^2}{256},S=-\frac{5ab^3}{2048}*$	A1*cso [4]
(c)	$-\frac{ab}{16}=\frac{128}{5}.\left(-\frac{5ab^3}{2048}\right)$ or $\frac{5}{128}.\left(-\frac{ab}{16}\right)=-\frac{5ab^3}{2048}$ or $-\frac{5ab}{2048}=-\frac{5ab^3}{2048}$ or $-\frac{ab}{16}=-\frac{ab^3}{16}$	M1
	$\left(b^2=1\right)b=1*$	A1*
	$\frac{3a\times1^2}{256}=\frac{9}{256}\Rightarrow a=3*$	B1*cso [3]
(d)	$\frac{\sqrt{6}}{2}=\frac{3}{2}\left(1+\frac{x}{4}\right)^{-\frac{1}{2}}$ or $\frac{\sqrt{6}}{2}=\frac{3}{\sqrt{4+x}}$ or $\frac{3}{\sqrt{6}}=\frac{3}{\sqrt{4+x}}$ or $\sqrt{6}=\sqrt{4+x}$ oe $\Rightarrow x=(2)$	M1
	$\frac{3}{2}-\frac{3\times1}{16}\times2+\frac{3\times3\times1^{(2)}}{256}\times2^2-\frac{5\times3\times1^{(3)}}{2048}\times2^3$	dM1
	1.207	A1 [3]
		Total 12 marks

Type MS · Using a binomial expansion to estimate a value

Answer 11 of 12 · Question 5 part b · 3 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

Number		
5 (a)	$1+\frac{1}{2}(-x)+\frac{\frac{1}{2}\left(-\frac{1}{2}\right)}{2!}(-x)^2+\frac{\frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{3!}(-x)^3$	M1
	$1-\frac{1}{2}x-\frac{1}{8}x^2-\frac{1}{16}x^3$	A1 A1 (3)
(b)	$x = 0.08$	B1
	$1-\frac{1}{2}"(0.08)"-\frac{1}{8}"(0.08)"^2-\frac{1}{16}"(0.08)"^3$	M1
	0.959168 cao	A1 cao (3)
(c)	$(\sqrt{0.92}=\frac{\sqrt{23}}{5}\Rightarrow\sqrt{23}=)"0.959168"\times 5$	M1
	4.79584	A1 (2) **[8]**

Part	Mark	Additional Guidance
(a)	M1	For an attempt at a Binomial expansion. An attempt is defined as the following • The expansion must start with 1 • The powers of their $-x$ must be correct • $-x$ must be used at least once • The denominators 2! And 3! must be seen. Accept 2 and 6 Can be implied by at least 2 correct terms in an expansion
	A1	For at least one term in x correct and fully simplified.
	A1	For the expansion fully correct and simplified. Ignore terms in higher powers of x.
(b)	B1	For finding the value of $x = 0.08$
	M1	For correctly substituting **their** value of x into the expansion **provided** $\lvert x\rvert < 1$ Use of **their expansion** or the correct expansion must be seen explicitly here
	A1	cao
(c)	M1	For use of their value from (b) in $\sqrt{0.92}=\frac{\sqrt{23}}{5}\Rightarrow\sqrt{23}="0.959168"\times 5$
	A1	Cao
		Allow the final M1 A1 if $\sqrt{0.92}\times 5$ is clearly written and 4.79584 is clearly indicated as the answer.

Type MS · Using a binomial expansion to estimate a value

Answer 11 of 12 · Question 5 part b · 3 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Using a binomial expansion to estimate a value

Answer 12 of 12 · Question 5 part c · 2 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

Number		
5 (a)	$1+\frac{1}{2}(-x)+\frac{\frac{1}{2}\left(-\frac{1}{2}\right)}{2!}(-x)^2+\frac{\frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{3!}(-x)^3$	M1
	$1-\frac{1}{2}x-\frac{1}{8}x^2-\frac{1}{16}x^3$	A1 A1 (3)
(b)	$x=0.08$	B1
	$1-\frac{1}{2}"(0.08)"-\frac{1}{8}"(0.08)"^2-\frac{1}{16}"(0.08)"^3$	M1
	0.959168 cao	A1 cao (3)
(c)	$(\sqrt{0.92}=\frac{\sqrt{23}}{5}\Rightarrow\sqrt{23}=)"0.959168"\times 5$	M1
	4.79584	A1 (2) **[8]**

Part	Mark	Additional Guidance
(a)	M1	For an attempt at a Binomial expansion. An attempt is defined as the following • The expansion must start with 1 • The powers of their $-x$ must be correct • $-x$ must be used at least once • The denominators 2! And 3! must be seen. Accept 2 and 6 Can be implied by at least 2 correct terms in an expansion
	A1	For at least one term in x correct and fully simplified.
	A1	For the expansion fully correct and simplified. Ignore terms in higher powers of x.
(b)	B1	For finding the value of $x = 0.08$
	M1	For correctly substituting **their** value of x into the expansion **provided** $\|x\| < 1$ Use of **their expansion** or the correct expansion must be seen explicitly here
	A1	cao
(c)	M1	For use of their value from (b) in $\sqrt{0.92}=\frac{\sqrt{23}}{5}\Rightarrow\sqrt{23}="0.959168"\times 5$
	A1	Cao
		Allow the final M1 A1 if $\sqrt{0.92}\times 5$ is clearly written and 4.79584 is clearly indicated as the answer.

Type MS · Using a binomial expansion to estimate a value

Answer 12 of 12 · Question 5 part c · 2 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks